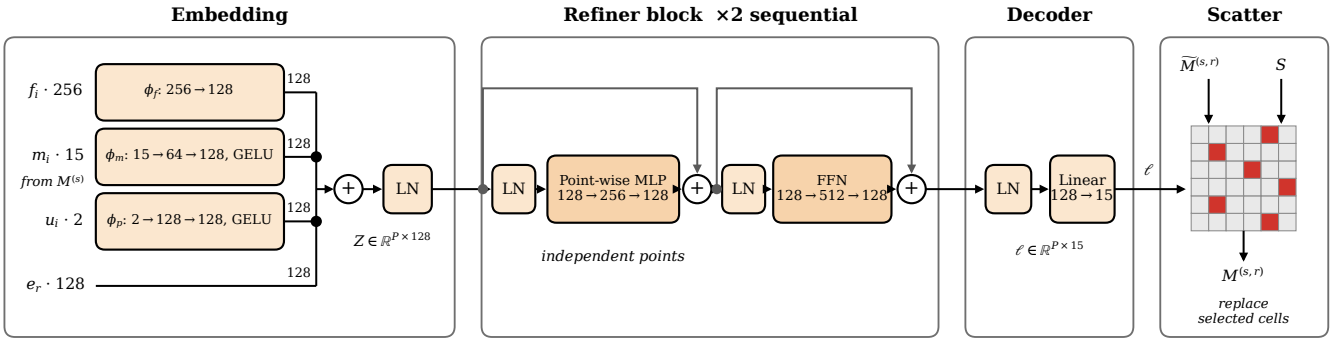
MLP point refiner
Embedding
$f_i \cdot 256$
ϕ_f: 256 → 128
128
$m_i \cdot 15$
from $M^{(s)}$
ϕ_m: 15 → 64 → 128, GELU
128
$u_i \cdot 2$
ϕ_p: 2 → 128 → 128, GELU
128
$e_r \cdot 128$
128
+
LN
$Z \in \mathbb{R}^{P \times 128}$
Refiner block ×2 sequential
LN
Point-wise MLP
128 → 256 → 128
+
LN
FFN
128 → 512 → 128
+
independent points
Decoder
LN
Linear
128 → 15
ℓ
$\ell \in \mathbb{R}^{P \times 15}$
Scatter
$\widetilde{M}^{(s,r)}$
S
$M^{(s,r)}$
replace
selected cells
$P = 196$ train / 784 test, per RoI and level